

Lecture 6: IoT System Design and Implementation

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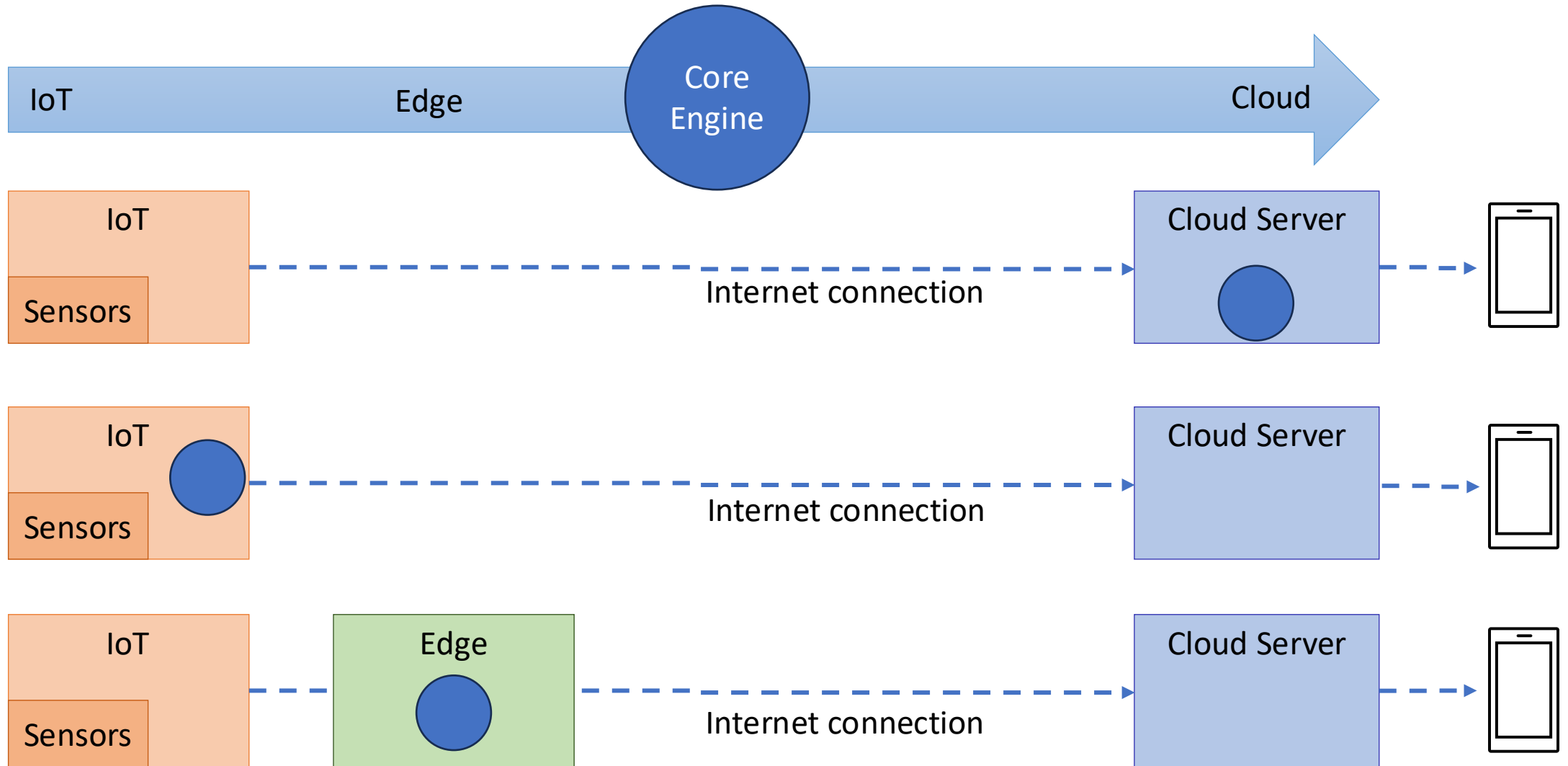
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Contents

- A systems perspective on sensing
 - How do we put it all together and build an end-to-end sensing system?
 - What are the missing blocks for building a real system based on what we have learned thus far?
- General pipeline
- Useful techniques
 - Windowing
 - Filtering
 - Target Detection
 - Peak Finding

System Architecture



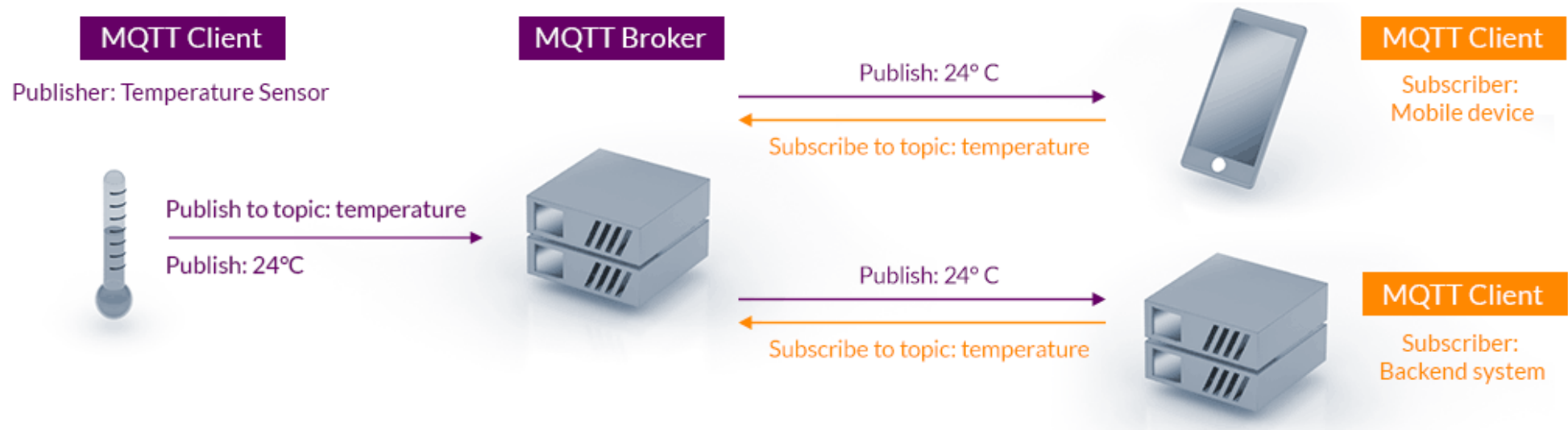
General Pipeline

- Hardware/sensor (cameras/WiFi/IMU/other sensors)
 - Standalone components? Plugged into computers?
 - Wired or wireless?
- Data Collection/Acquisition
 - How to fetch data from certain sensors from the host machine?
 - How often/how much to fetch?
 - What communication technology & network protocol?
- Preprocessing
- Core Algorithms
- User-End Interface



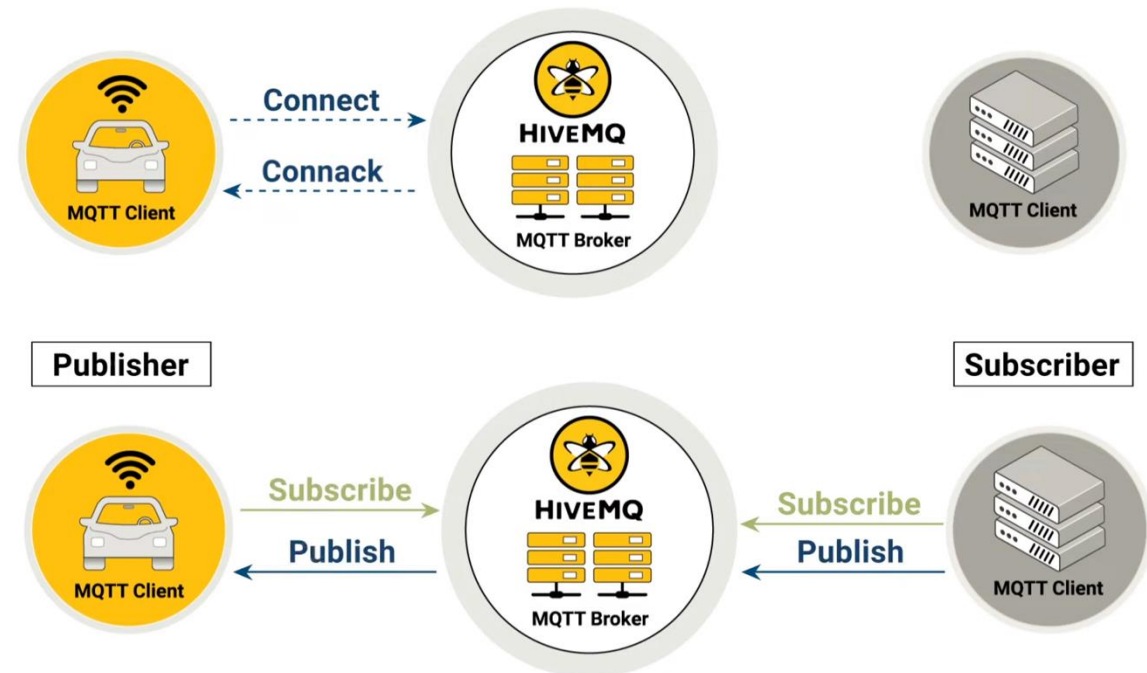
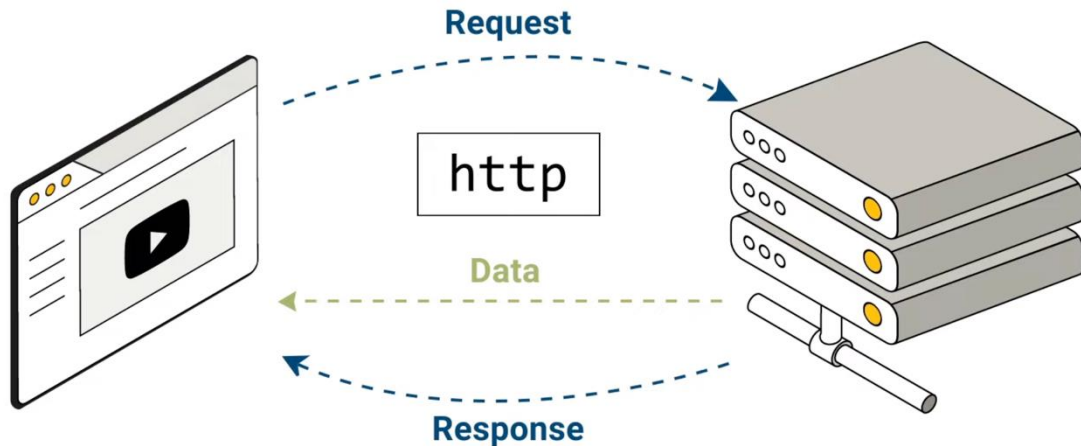
Data Transportation: MQTT

- MQTT: The Standard for IoT Messaging (source: AWS)
 - Simple implementation
 - Lightweight and efficient
 - Scalable
- MQTT (Message Queuing Telemetry Transport)



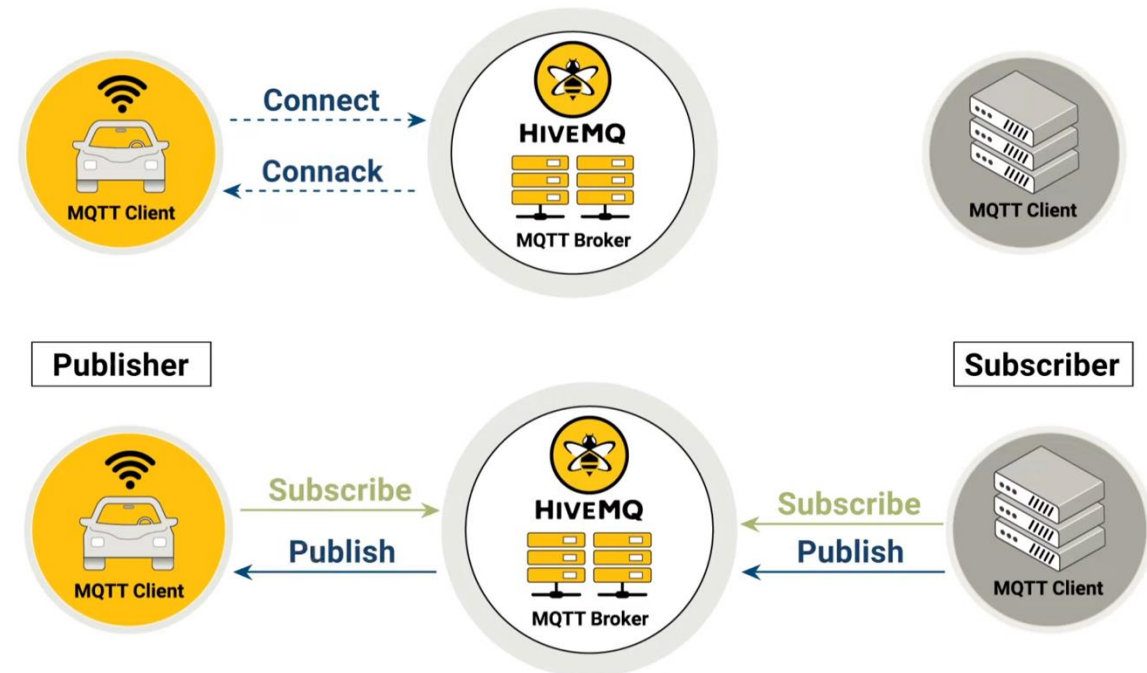
MQTT Architecture

- Publish/Subscribe model (Producer/Consumer)
- Components
 - Client, Broker, Connection



MQTT Architecture

- Publish/Subscribe model (Producer/Consumer)
- Components
 - Client, Broker, Connection
- Scalability
 - Space decoupling
 - Time decoupling
 - Synchronization decoupling



MQTT Topics

Topics are structured in a hierarchy like folders and files in a file system
Allow to create a user friendly and self descriptive naming structures

myhome / groundfloor / livingroom / temperature
topic level topic level

‘+’ matches anything at a given tree level

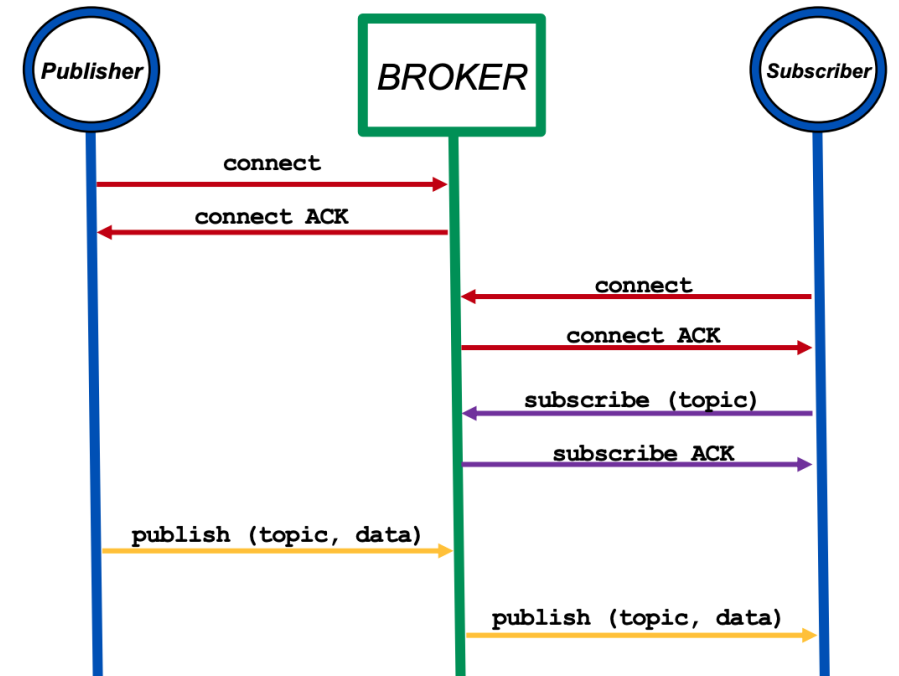
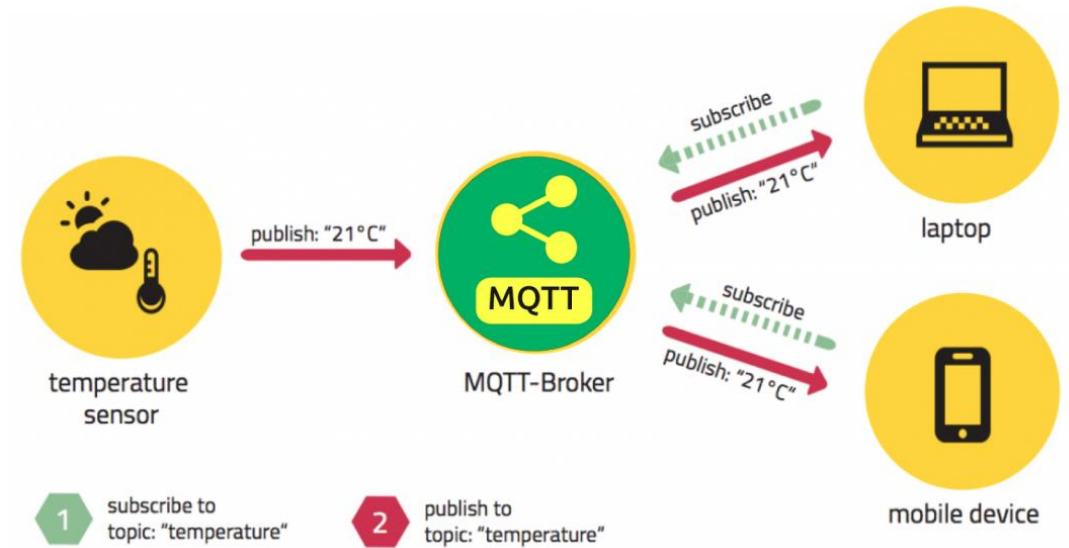
myhome / groundfloor / + / temperature

- ✓ myhome / groundfloor / livingroom / temperature
- ✓ myhome / groundfloor / kitchen / temperature
- ✗ myhome / groundfloor / kitchen / brightness
- ✗ myhome / firstfloor / kitchen / temperature
- ✗ myhome / groundfloor / kitchen / fridge / temperature

‘#’ matches a whole sub-tree

myhome / groundfloor / #
only at the end
multiple topic levels

- ✓ myhome / groundfloor / livingroom / temperature
- ✓ myhome / groundfloor / kitchen / temperature
- ✓ myhome / groundfloor / kitchen / brightness
- ✗ myhome / firstfloor / kitchen / temperature



MQTT Resources

- <http://mqtt.org/>
- The most widely used implementations
 - <http://mosquitto.org/>
 - <https://www.hivemq.com/>
- The MQTT community wiki:
<https://github.com/mqtt/mqtt.github.io/wiki>
- A good tutorial: <http://www.hivemq.com/mqtt-essentials/>

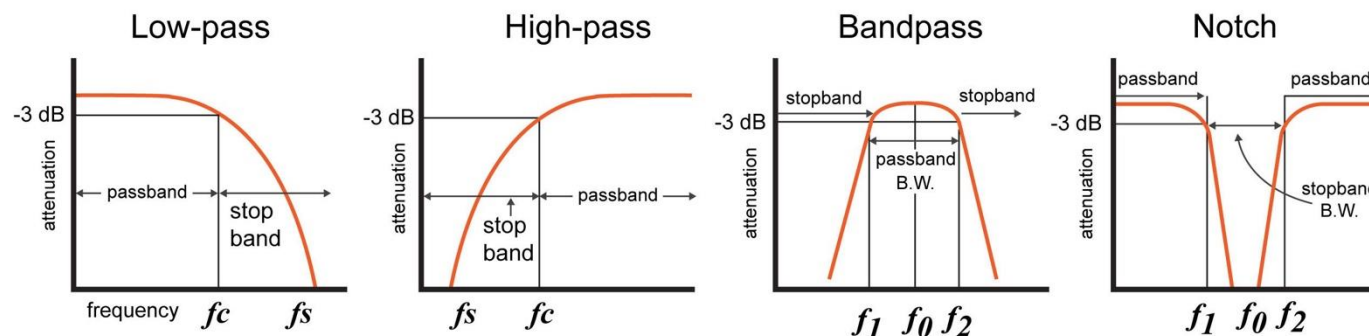
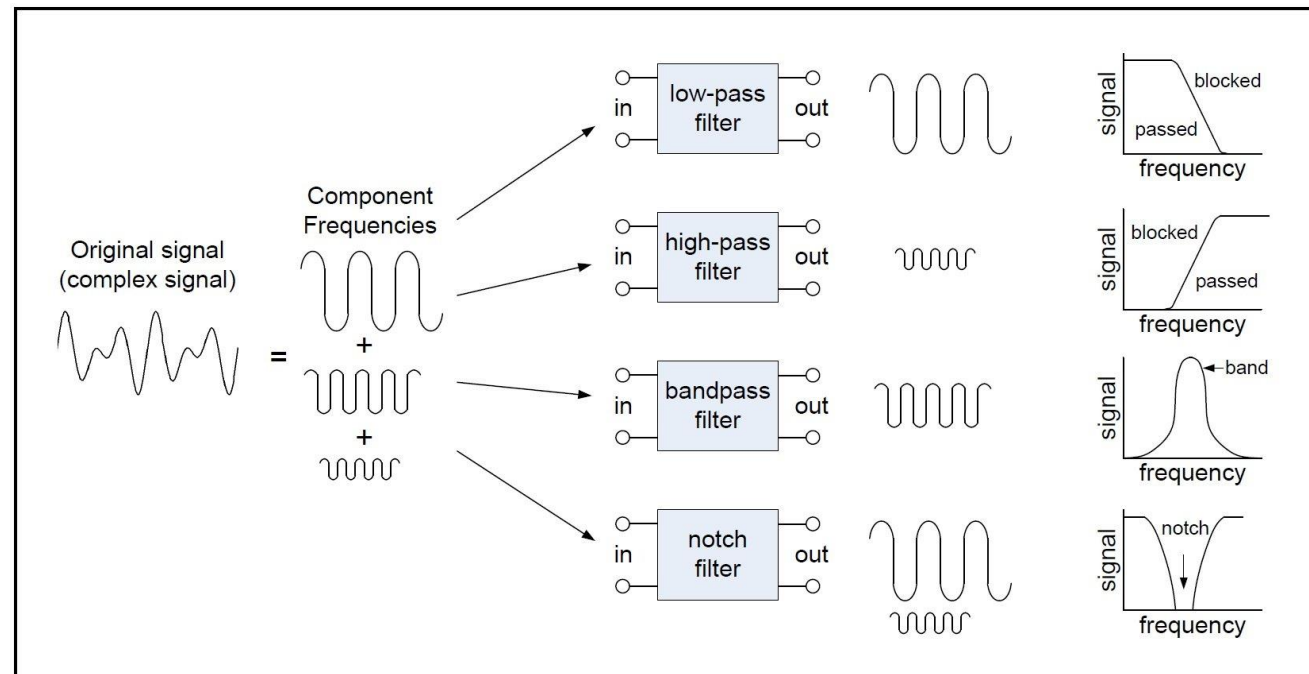


Windowing

- Very often, we are processing time-series data.
- Window length
 - Any fundamental limits? E.g., FFT window requirement
 - Trade-off of accuracy vs. latency
 - Other constraints: Sampling rate, size, communication capacity, etc.
- Sliding step
 - Computation vs. responsiveness



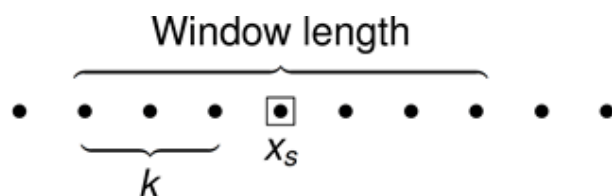
- Low-pass filter
- High-pass filter
- Band-pass filter
- Band-stop (Notch) filter



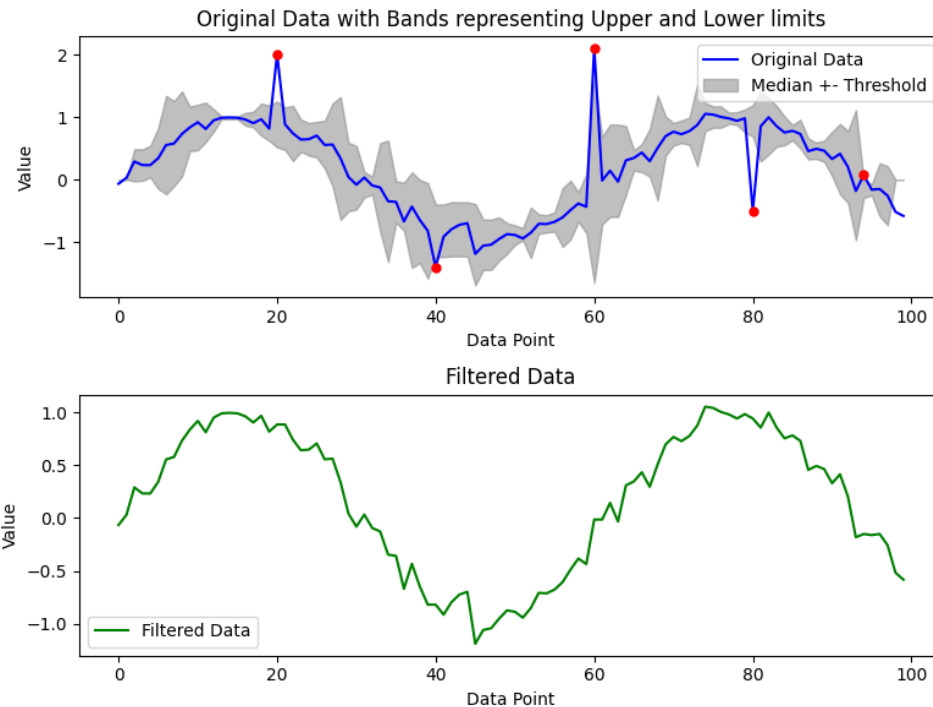
Filtering

- Hampel filter

- A robust method for outlier detection
- Median Absolute Deviation (MAD)



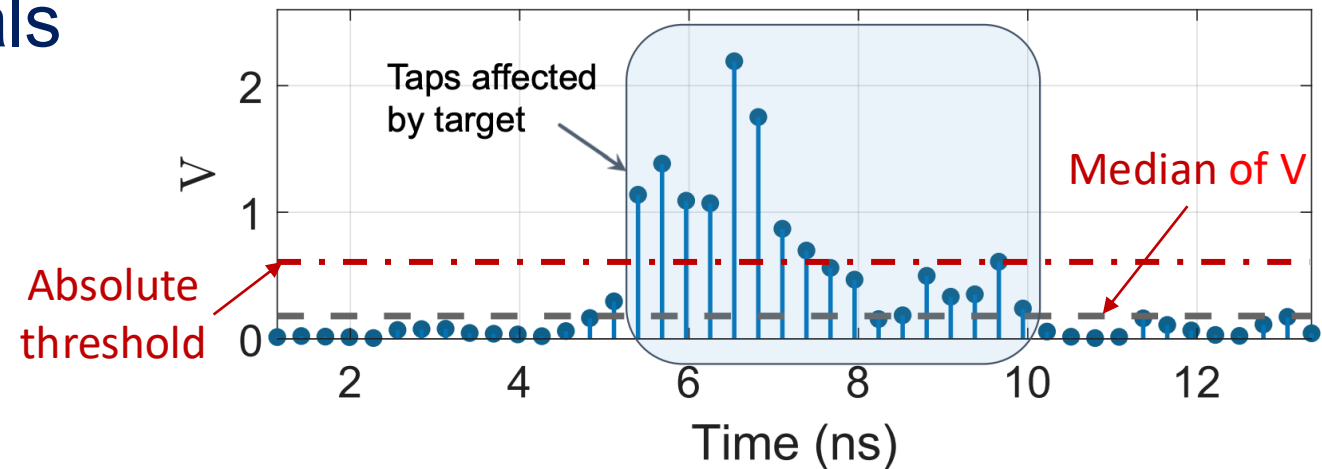
- Set the current window (of odd length)
- Get the local median m and SD σ
- Outlier if $|x_s - m_i| > n\sigma$, $n := \text{threshold}$



https://github.com/MichaelisTrofficus/hampel_filter

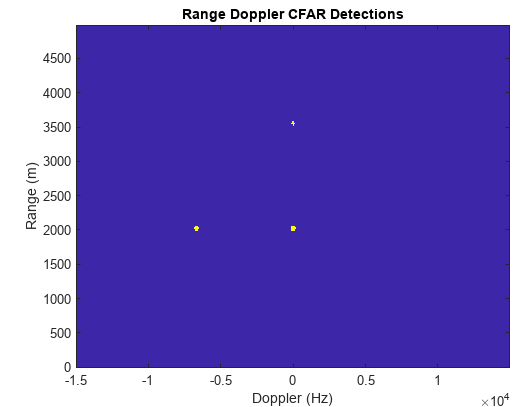
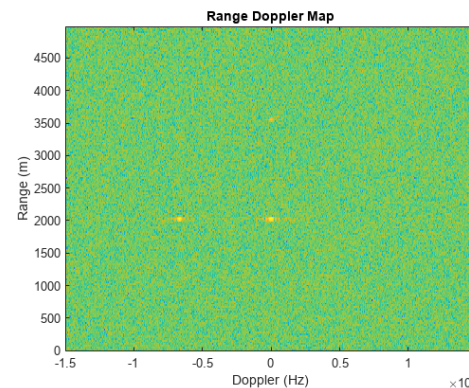
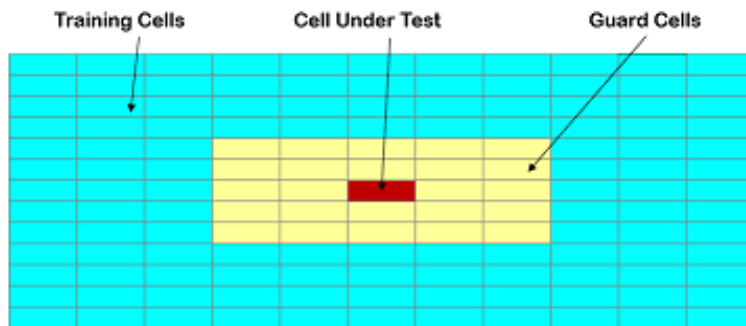
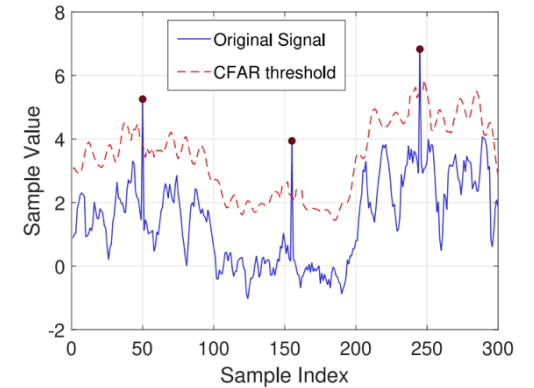
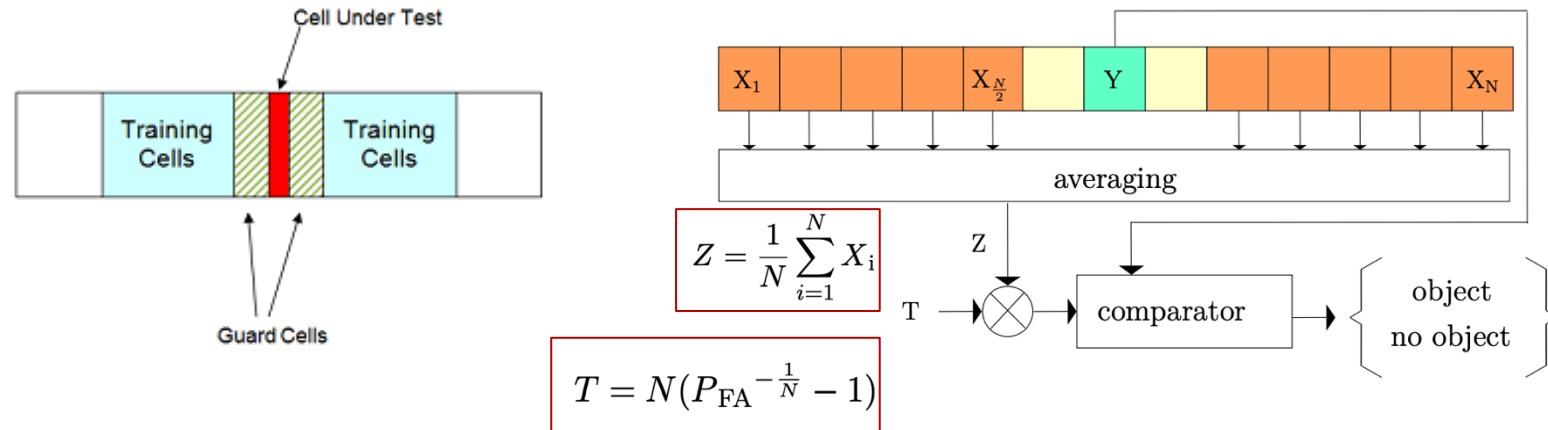
Detection: Thresholding

- Problem: How to detect a target/signal/event of interest?
- Hard thresholding
 - Empirical values
- Median filter for sparse signals



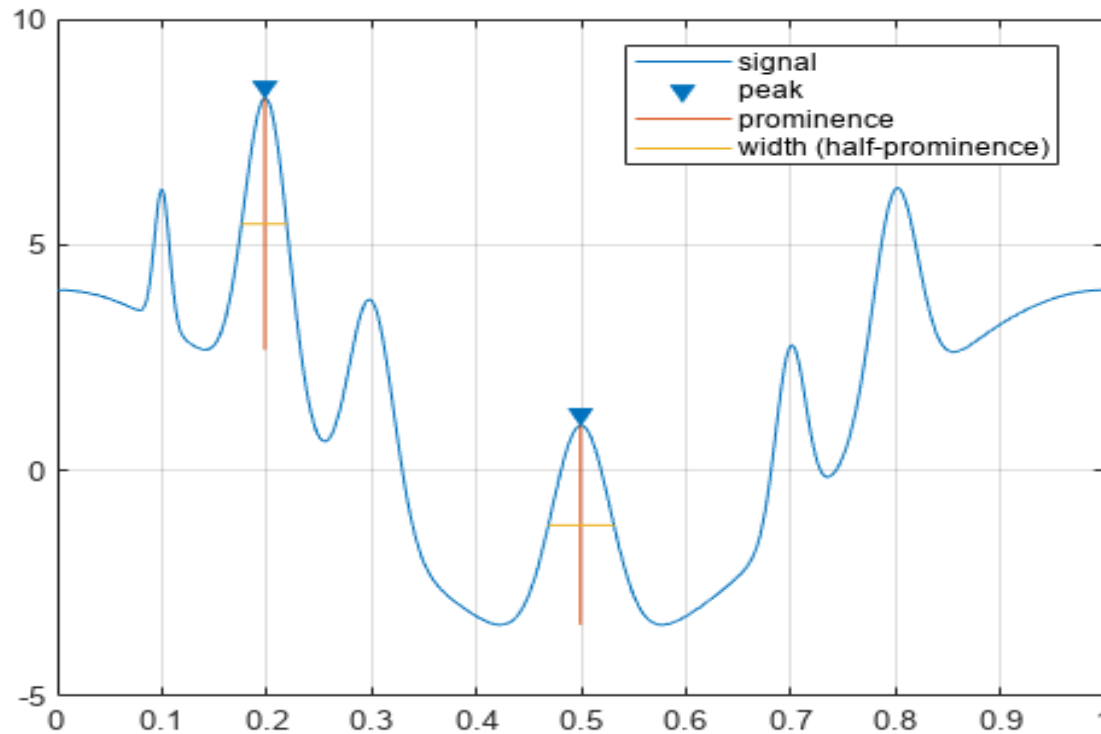
Detection: CFAR

- Constant False Alarm Rate
 - Cell-Averaging CFAR (CA-CFAR)

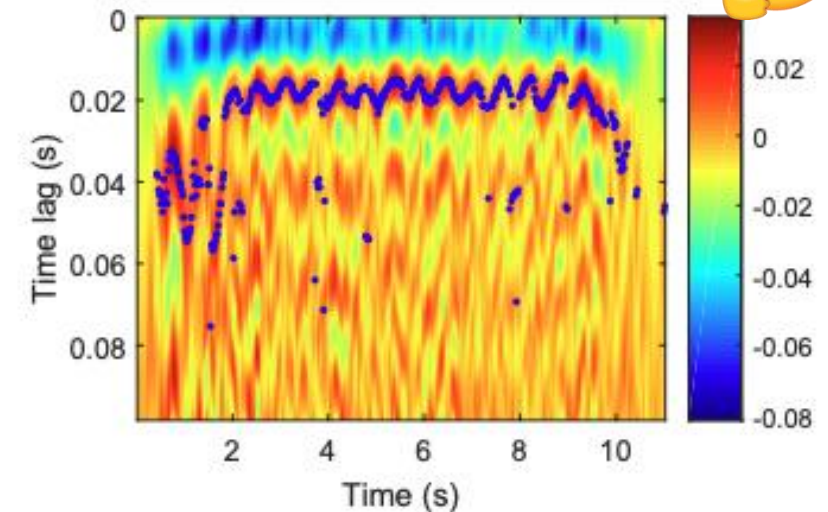


Detection: Peak

- `findpeaks(S, 'MinPeakProminence', 4, 'Annotate', 'extent')`



How can you make peak finding more accurate and robust for a time series?



A Working Example

- We learn how motion detection works. How to build a system?
 - Suppose we plan to build a system using ESP32 WiFi IoT.

An Advanced Working Example

- Building a tech solution for Wilderness Search and Rescue (WiSAR) using Wi-Fi on drones and smartphones
 - How do we design, implement, and evaluate the system?
 - Will be back to this after Lec08 on Localization. Stay tuned^_^

Questions?

- Thank you!